

# Christian Rafael de Souza Silva

Software Engineer — ML Systems, AI Infrastructure & Systems Software

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15 years of software development experience across enterprise systems, APIs, web and desktop applications, with additional focus on ML systems, AI infrastructure, low-level systems and data-driven products.



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## EXPERIENCE

Nov 2023 — Present

### Information Systems Programmer — ITB Equipamentos Elétricos

Birigui, SP, Brazil

- Internal management system covering finance, accounting, production planning and engineering.
- C++, MySQL and Delphi; Horse APIs and Swagger documentation.
- .NET Core, EF Core, C#, JavaScript, React and PostgreSQL.
- DDD, CQRS, layered architecture and automated testing.
- AI-assisted agile development and Spec Driven Development practices.

Feb 2016 — Nov 2023

### Full Stack Developer — Gigatron Franchising / Dotbank

Birigui, SP, Brazil

- PHP, Laravel, Angular, Vue and PostgreSQL; Android apps in Java/Kotlin and hybrid React/Ionic apps.
- APIs and products for ZIP code lookup, SMS, food delivery, retail automation, fiscal documents and ticketing.
- Linux/Windows Server administration; DigitalOcean, AWS and Cloudflare.
- Delphi application for production sheet stacking and printing.

Jul 2014 — Feb 2016

### Full Stack Developer — Netstart Informática

Birigui, SP, Brazil

- Retail automation with Delphi/PostgreSQL and SAT fiscal queue processing with Redis.
- Desktop updater; PHP/CodeIgniter API and panel with jQuery and PostgreSQL.
- Internal tools, Bash scripts and cloud storage/querying of fiscal documents.

Jan 2014 — Jul 2014

### Full Stack Developer — Gigatron Franchising

Birigui, SP, Brazil

- PHP/MySQL applications, server administration and SMS service using Python with a modem.

Aug 2012 — Jun 2013

### Programmer Apprentice — Assessor Público (MV & P Tecnologia)

Araçatuba, SP, Brazil

- GeneXus applications, R&D, AWS server administration and OpenVPN.

Jan 2012 — Jul 2012

### Programmer Apprentice — Gigatron Tecnologia

Birigui, SP, Brazil

- Internal applications in Delphi.

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## SELECTED PROJECTS

2025 —

### ChrisOS

Experimental operating system and low-level playground for kernels, compilers, toolchains, virtual machines, emulators, drivers and graphics subsystems. [GitHub](#) · [Documentation](#)

2026

### Chris-GPT-2

124M-parameter GPT-2 trained from scratch on ~10B FineWeb-Edu tokens using 4× A100 GPUs, with SafeTensors/GGUF exports. [GitHub](#) · [Hugging Face](#)

2026 —

### **Chris-Llama**

Experimental LLM inference runtime in C covering GGUF, tokenization, attention, KV cache, quantization and CPU/GPU execution. [GitHub](#)

2026 —

### **Chris-Torch**

Deep learning framework built from scratch with tensors, autograd, neural-network components, optimizers and accelerated backends. [GitHub](#)

2026 —

### **Velis**

Local-first collaborative workspace built with Next.js, NestJS, PostgreSQL and Redis; implemented offline mutation replay, editor convergence logic, WebSocket synchronization and granular access control. [Product](#)

2026 —

### **Chris Cleaner**

C#/ .NET desktop disk-analysis utility with Avalonia UI, Win32-native file enumeration, NTFS USN Journal monitoring, resumable indexing and a background Windows service. [Product](#)

2026 —

### **Engineering Manual**

Open, bilingual computer science and software engineering reference with full-text search, structured study tracks, editorial checks and automated static publishing. [Website](#) · [GitHub](#)

2026 —

### **SharpEx**

Market and portfolio-intelligence workspace with multiple portfolios, risk/return metrics, crypto, market data and structural research integration.

2026 —

### **Marginal Thinking**

Independent macroeconomics, markets and geopolitics research platform with recurring reports, structural datasets and primary-source analysis. [marginalthinking.org](#)

## **EDUCATION**

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- **Postgraduate Degree in Electronic Engineering & Robotics** — UNINTER, 2026
- **Associate Degree in Data Science** — UNINTER, 2023 — 2025
- **Bachelor's Degree in Electrical Engineering** — UNINTER, 2025 — 2030, in progress
- **Technical Degree in Mechatronics** — Colégio Salesiano Dom Lasagna, 2012 — 2014

## **TECHNICAL PROFILE**

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**Languages:** C, C++, Python, Rust, C#, Java, Kotlin, JavaScript/TypeScript, PHP, Delphi.

**ML & systems:** LLM training/inference, PyTorch, CUDA, distributed training, GGUF, quantization, Linux, operating systems, compilers, virtual machines and emulators.

**Applications & infrastructure:** .NET, React, Laravel, REST APIs, PostgreSQL, MySQL, Redis, DDD, CQRS, GitHub Actions, Cloudflare, AWS and DigitalOcean.